

LankaJourney

Smart Sri Lankan Travel Planning System

Project Report | COM1305 | DevSphere

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1. Project Overview

LankaJourney is a travel recommendation system designed to help travelers plan a suitable Sri Lankan journey within their budget, time limits, comfort requirements, and destination preferences. The system evaluates personal preferences such as budget, number of days, adventure level, weather, and transport mode, then suggests a travel route using available destinations, accommodation, restaurants, and transport options.

2. Purpose and Problem Addressed

Many tourists struggle to combine destinations, accommodation, transport, and daily activities into a well-balanced plan. LankaJourney solves this by creating a structured decision-support system that matches a traveler profile with suitable locations, services, and routes across Sri Lanka.

3. Objectives

- ? Collect traveler information including budget, duration, destination preferences, weather tolerance, comfort level, and transportation preference.
- ? Recommend destinations and activities that match the traveler profile.
- ? Suggest suitable accommodation and restaurant choices based on price, rating, comfort, and food availability.
- ? Store and organize travel-related data using a structured MySQL database.
- ? Generate a travel route using a scoring logic based on compatibility, accessibility, and travel time.

4. Core Functional Requirements

- ? Traveler profile inputs: budget, trip duration, preferred regions, journey style, physical activity level, weather preference, and comfort level.
- ? Accommodation details: accommodation ID, name, type, location, price, rating, luxury level, food availability, and contact information.
- ? Tourist place details: place ID, name, type, location, visit duration, difficulty, accessibility, rating, and description.
- ? Transportation options: private vehicle, public bus, train, or mixed transportation preferences; rental details include passenger count, luggage, and driver requirement.
- ? Food suggestions: restaurants and dining options when accommodation does not include food facilities.
- ? Route generation: compute the best travel combination using criteria such as budget compatibility, travel time, accommodation suitability, and destination preferences.

5. System Design and Data Structure

Module	Description
Location	Stores coordinates, town, district, and province information for destinations.
Accommodation	Stores accommodation type, cost, rating, food availability, and location details.
Tourist_Places	Contains attractions, categories, climate, accessibility, and difficulty information.
Vehicle_Agency	Stores rental agencies and their contact information.
Vehicle_Rent	Stores fleet options, passenger count, luggage capacity, and driver requirement.
Restaurant	Stores restaurant details, rating, and contact data.

Database notes:

The project database is based on the LankaStay schema with records for locations, accommodation providers, tourist destinations, vehicle agencies, rental options, and restaurants. The sample dataset includes 110 locations, 200 accommodations, 200 tourist sites, 20 agencies, 100 rental options, and 200 restaurants.

6. User Interface and Implementation

The UI is implemented in C++ using Windows Forms in the LankaJourneyUI project. The dashboard includes navigation elements such as Dashboard, Planning, Accommodations, Destinations, Vehicles, Bookings, and Settings. The design provides a clean, user-friendly entry point for travelers to browse trip information and launch planning activities.

7. Route-Generation Logic

The system uses a scoring approach to calculate the best combination of destinations and services. The major factors are budget compatibility, destination preferences, weather compatibility, activity compatibility, travel time, accommodation suitability, and transport compatibility. This logic allows the application to prioritize journeys that align with the traveler profile and available resources.

8. Expected Benefits

- ? Helps travelers create personalized Sri Lanka tours within their budget and time.
- ? Improves decision-making by combining destination, accommodation, restaurant, and transport information in one system.
- ? Supports trip planning for different travel styles such as relaxing, adventure, cultural, wildlife, beach, and family trips.
- ? Provides a structured database that can be expanded with new destinations and travel service providers.

9. Conclusion

LankaJourney is a practical travel-planning solution that demonstrates how data-driven recommendation systems can simplify itinerary creation. By combining traveler preferences with structured destination and accommodation data, the project provides a foundation for an intelligent travel assistant for Sri Lanka. Future versions can extend the system with live map integration, booking APIs, and web/mobile deployment.

10. Future Enhancements

- ? Integrate Google Maps or OpenStreetMap APIs for real-time routing and distance calculations.
- ? Add hotel booking, vehicle booking, and restaurant reservation features.
- ? Create a web/mobile version with login and user profile storage.
- ? Introduce machine-learning-based recommendations based on traveler behavior and historical preferences.